

Narendran Palanivel
Data Analyst

Location: TX | narendan0510@gmail.com | 682-382-2206 | https://www.linkedin.com/in/narendran-palanivel-b922aa190/

PROFESSIONAL SUMMARY

- Results-driven Data Analyst with 3 year’s experience delivering end-to-end data solutions across pharmaceutical, fintech, and advertising sectors, specializing in data integration, analytics, and automation.
- Expert in ETL pipeline development using Informatica, AWS Glue, Azure Data Factory, Apache Airflow, and DBT, streamlining multi-source data ingestion and transformation for high-volume datasets.
- Advanced analytics & machine learning expertise in Python, SQL, and R, leveraging Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn, Plotly, NLTK, and Statsmodels for regression, classification, anomaly detection, and fraud prevention models.
- Strong data visualization skills with Power BI, Tableau, Microsoft Excel advanced charts and pivot tables, and Plotly, creating interactive dashboards and automated reports to drive strategic insights.
- Cloud & DevOps proficiency across AWS S3, Lambda, RDS, Redshift, Azure, and Google Cloud BigQuery, with hands-on experience in Jenkins, Docker, Terraform, Git, and GitHub for scalable, automated, Agile-driven data delivery.
- Extensive database expertise with MySQL, PostgreSQL, Oracle, MongoDB, SQLite, and Snowflake, applying SQL-based ETL pipelines and data warehousing Star and Snowflake schema.
- Project achievements include credit risk scoring, drone-based YOLOv9 surveillance with real-time Kafka streaming, and IoT-enabled smart home monitoring published by Springer, showcasing technical innovation and real-world problem-solving.

EDUCATION

The University of Texas	Arlington, TX, USA
Master of Science in Data Science	August 2023 - May 2025
Amrita Vishwa Vidyapeetham	India
Bachelor’s of Technology in Electricals and Electronics Engineering	Jun 2017 - May 2021

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, HTML, JavaScript, R, Shell Scripting
Python Libraries & Frameworks: Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn, Plotly, NLTK, Statsmodels
Databases: MySQL, PostgreSQL, Oracle, MongoDB, SQLite, Snowflake
Data Visualization Tools: Power BI, Tableau, Microsoft Excel (Advanced Charts, Pivot Tables), Plotly
Analytics & Statistical Skills: Exploratory Data Analysis (EDA), Data Mining, Data Cleaning, Feature Engineering, Hypothesis Testing, A/B Testing, Time Series Analysis, Text Mining, Predictive Modeling
ETL & Data Engineering: Apache Airflow, AWS Glue, Informatica, DBT, Data Warehousing (Star/Snowflake Schema), SQL-based ETL pipelines
Cloud Platforms & DevOps Tools: AWS (S3, Lambda, RDS, Redshift), Google Cloud (BigQuery), Azure, Jenkins, Docker, Git, GitHub, Terraform
APIs & Scripting: REST API Integration, Web Scraping, JSON/XML Parsing, Automation with Python Scripts
Tools & IDEs : Google Colab, Jupyter Notebook, Visual Studio Code, PyCharm, GitHub
Operating Systems: Windows, MacOS, Linux/Unix

PROFESSIONAL EXPERIENCE

COGNIZANT	India
Programmer Analyst	Jun 2021– Aug 2023
Technologies Used: Python, SQL, Informatica PowerCenter, Tableau, PowerBI, Microsoft Excel (Pivot Tables, VLOOKUP, Conditional Formatting), AWS S3, RDS, Redshift, Agile/Scrum, Git, Jenkins	
Directed Master Data Management (MDM) and data integration projects for Abbott Pharmaceuticals, ensuring enterprise-wide data consistency and 100% regulatory compliance.	
Developed and optimized Informatica & SQL-based ETL pipelines, integrating multi-source datasets, while automating Python and SQL validations for 100K+ credit card applications cutting processing time by 20+ hours/month, achieving 98% accuracy, boosting fraud detection, and reducing manual reviews by 25%.	
Enhanced and maintained 50+ high-performance data workflows, improving accuracy by 15% and streamlining operational processes.	
Collaborated with cross-functional teams to design and manage AWS S3 for secure dataset storage, RDS for transactional operations, and Redshift for analytical workloads, assisting in query optimization, performance improvement, and scalable solution development.	
Redesigned error-handling frameworks and implemented data governance strategies, lowering system-related rejections by 20%, improving audit scores by 15 points, and reducing regulatory risk.	
Delivered Tableau and Power BI dashboards in Agile sprints, accelerating reporting cycles by 30% and improving visibility into business KPIs.	
Executed continuous data cleansing, validation, and integration, ensuring high-quality inputs for analytics and visualization tools.	
Created Excel-based performance and exception reports using pivot tables, VLOOKUP, advanced formulas, and conditional formatting, enabling faster decision-making and root cause analysis.	

Data Analyst Intern

Dec 2020– May 2021

Technologies Used: Python, SQL, Excel (Pivot Tables, VLOOKUP, Conditional Formatting), Tableau, Azure Data Lake, Azure SQL Database, Azure Synapse Analytics, Agile/Scrum

- Designed SQL-based ETL pipelines across MySQL and Snowflake, processing 500K+ records and cutting query response time by 10 hours/week.
- Automated Python (Pandas, NumPy) data preprocessing, reducing manual cleaning time from 3 hours to under 20 minutes.
- Developed interactive Tableau and Power BI dashboards, accelerating campaign launches by 2 days for marketing teams.
- Analyzed customer behavior using R and Seaborn EDA, driving targeted retention strategies.
- Integrated real-time survey data via REST APIs and Python scripts, ensuring up-to-date insights for weekly operations reviews.
- Configured Azure Data Lake, Azure SQL Database, and Azure Synapse Analytics, enabling secure storage, faster queries, and optimized workflows.
- Built a Scikit-learn logistic regression model predicting customer churn, preventing \$7,500 in acquisition costs.
- Created Excel-based KPI trackers with pivot tables, VLOOKUP, and dynamic charts, saving 6+ hours/week in manual reporting.

PROJECT AND LEADERSHIP

Autonomous Drone Navigation and Surveillance System

Role: Lead Developer | **Output:** Automated aerial surveillance with >90% anomaly detection accuracy

- Designed a drone-based surveillance system using Apache Kafka for real-time streaming of video and sensor data.
- Applied YOLOv9 for object detection, achieving >90% accuracy in identifying humans, vehicles, and anomalies in aerial feeds.
- Automated drone navigation using Python & Tello Py, enabling autonomous takeoff, path control, and video capture.
- Integrated reinforcement learning algorithms for dynamic route optimization, reducing flight energy consumption by 15%.
- Built a scalable edge-to-cloud pipeline for real-time anomaly detection, improving situational awareness for security teams.

Technologies Used: Python, Apache Kafka, YOLOv9, Tello Py, Reinforcement Learning, OpenCV, AWS, Edge Computing

Smart Home with Condition Monitoring

Role: IoT & Data Pipeline Engineer | **Output:** Improved safety response times by 40%

- Developed an IoT-enabled smart home system with embedded sensors for real-time monitoring of temperature, motion, and occupant health.
- Built Python & SQL pipelines for continuous data processing, enabling instant alerts and automated responses.
- Created interactive dashboards to visualize environmental trends and safety metrics from remote interfaces.
- Applied predictive analytics models to detect abnormal patterns, preventing safety risks proactively.
- Optimized multi-sensor data ingestion, achieving sub-second response times for critical alerts.

Technologies Used: Python, SQL, IoT Sensors, Power BI, Predictive Analytics, AWS IoT, MQTT, Data Warehousing

Credit Card System – Data Integration Project

Role: ETL Developer | **Output:** 100% compliance in credit card application validation workflows

- Designed and deployed Informatica PowerCenter ETL pipelines to consolidate customer and transaction data from multiple sources into a centralized data warehouse.
- Built modular workflows for customer verification and bank account validation using lookup transformations, SQL-based filters, and conditional logic.
- Implemented eligibility rule checks and dynamic mappings to validate applicant history, including credit card ownership and account status.
- Enhanced data reliability through error-handling, audit trail logging, and monitoring, increasing process reliability by 25%.

Technologies Used: Informatica, PowerCenter, SQL, Oracle, Data Warehousing, ETL Automation, Excel, Git